

Extending the preconditioning approach for low-Mach Navier-Stokes equations: General Approach

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1 Energy equation for reacting flows

For reacting flows, we have the energy equation for the enthalpy h [3] (sensible + chemical) given by:

$$\frac{\partial \rho h}{\partial t} + \nabla \cdot (\rho \vec{u} h) = \underbrace{\frac{Dp}{Dt}}_{\text{term 1}} + \nabla \cdot (\kappa \nabla T) - \nabla \cdot \left(\rho \sum_{i=1}^N h_i Y_i \vec{V}_i \right) + \underbrace{\underline{\tau} : \nabla \vec{u}}_{\text{term 2}} + \underbrace{\dot{Q}}_{\text{term 3}} + \underbrace{\rho \sum_{i=1}^N Y_i \vec{f}_i \vec{V}_i}_{\text{term 4}} \quad (1)$$

where the enthalpy h is given by:

$$h = \underbrace{\int_{T_0}^T C_p dT}_{\text{sensible}} + \underbrace{\sum_{i=1}^N \Delta h_{f,i}^0 Y_i}_{\text{chemical}} \quad (2)$$

here T is the temperature of the gas mixture, T_0 is a reference temperature usually set to 298.15 K, $h_{f,i}^0$ is the enthalpy of formation of species i at T_0 , and the specific heat capacity at a constant pressure of the gas mixture C_p is given by:

$$C_p = \sum_{i=1}^N C_{p,i} Y_i \quad (3)$$

where $C_{p,i}$ and Y_i are the specific heat capacity at a constant pressure of species i , and mass fraction of species i , respectively.

Additionally, the enthalpy for species i , h_i , is defined by:

$$h_i = \int_{T_0}^T C_{p,i} dT + \Delta h_{f,i}^0 \quad (4)$$

using (3), it follows that:

$$h = \sum_{i=1}^N Y_i h_i \quad (5)$$

Finally, ρ , $\vec{u}$, p , $\underline{\tau}$, κ , and $\vec{V}_i$ are the density, velocity field, pressure, shear stress tensor, thermal conductivity, and diffusion velocity of species i , respectively. Moreover, $\dot{Q}$ and $\vec{f}_i$ are the volumetric heat sources and body forces on species i .

For low Mach number flows, the pressure term vanishes (term 1 in (1)), viscous heating is often assumed negligible (term 2 in (1)), body forces can be assumed zero (term 4 in (1)), and we can assume that there are no volumetric heat sources (term 3 in (1)). Then, equation (1) simplifies to:

$$\frac{\partial \rho h}{\partial t} + \nabla \cdot (\rho \vec{u} h) = \nabla \cdot (\kappa \nabla T) - \nabla \cdot \left(\rho \sum_{i=1}^N h_i Y_i \vec{V}_i \right) \quad (6)$$

assuming the diffusion velocity $\vec{V}_i$ given by:

$$Y_i \vec{V}_i = -D_i \nabla Y_i \quad (7)$$

equation (6) can be written as:

$$\frac{\partial \rho h}{\partial t} + \nabla \cdot (\rho \vec{u} h) = \nabla \cdot (\kappa \nabla T) + \nabla \cdot \left(\rho \sum_{i=1}^N h_i D_i \nabla Y_i \right) \quad (8)$$

2 Energy equation for multicomponent non-reacting flows

Following the derivations presented in section 1, the sensible enthalpy h_s [3] can be defined as follows:

$$h_s = \int_{T_0}^T C_p dT \quad (9)$$

Similarly, the sensible enthalpy of species i , $h_{s,i}$, is given by:

$$h_{s,i} = \int_{T_0}^T C_{p,i} dT \quad (10)$$

Then, combining equations (9) and (3), we obtain the following relation:

$$h_s = \sum_{i=1}^N Y_i h_{s,i} \quad (11)$$

Using the definitions provided in (9) and (10), equation (8) can be rewritten in terms of the sensible enthalpy as follows:

$$\frac{\partial \rho h_s}{\partial t} + \nabla \cdot (\rho \vec{u} h_s) = \nabla \cdot (\kappa \nabla T) + \nabla \cdot \left(\rho \sum_{i=1}^N h_{s,i} D_i \nabla Y_i \right) - \sum_{i=1}^N \Delta h_{f,i}^0 \left(\frac{\partial (\rho Y_i)}{\partial t} + \nabla \cdot (\rho \vec{u} Y_i) - \nabla \cdot (\rho D_i \nabla Y_i) \right) \quad (12)$$

Considering the transport equation for species i is given by:

$$\frac{\partial (\rho Y_i)}{\partial t} + \nabla \cdot (\rho \vec{u} Y_i) = \nabla \cdot (\rho D_i \nabla Y_i) + \dot{\omega}_i \quad (13)$$

where $\dot{\omega}_i$ is the reaction rate of species i (for more information, see section 12), we can substitute (13) into 12 to derive an energy equation for the sensible enthalpy given by:

$$\frac{\partial \rho h_s}{\partial t} + \nabla \cdot (\rho \vec{u} h_s) = \dot{\omega}_T + \nabla \cdot (\kappa \nabla T) + \nabla \cdot \left(\rho \sum_{i=1}^N h_{s,i} D_i \nabla Y_i \right) \quad (14)$$

where $\dot{\omega}_T$ is the heat release due to combustion given by:

$$\dot{\omega}_T = - \sum_{i=1}^N \Delta h_{f,i}^0 \dot{\omega}_i \quad (15)$$

It is important to note that, for non-reacting flows, the heat release due to combustion (Equation (15)) becomes zero. Therefore, the energy equation for the sensible enthalpy is more appropriate for multicomponent non-reacting flows, as it does not require the computation of the chemical enthalpy.

3 Energy equation for a perfect gas

In the current implementation of SU2, the energy equation for incompressible flows is given as follows [1]:

$$\frac{\partial (C_p \rho T)}{\partial t} + \nabla \cdot (C_p \rho T \vec{u}) = \nabla \cdot (\kappa \nabla T) \quad (16)$$

4 Energy equation for low-mach number flows: General form

Considering equations (8), (14), and (16) (and potentially other energy equations), the energy equation for low-Mach number flows can be expressed as:

$$\frac{\partial (\rho E)}{\partial t} + \nabla \cdot (\rho E \vec{u}) = \nabla \cdot F^v (T, \nabla T) + Q \quad (17)$$

where

$$E = \begin{cases} C_p T & , \quad \text{perfect gas} \\ h_s & , \quad \text{real gas sensible enthalpy} \\ h & , \quad \text{real gas enthalpy} \end{cases} \quad (18)$$

and F^v and Q are the viscous and source terms, respectively, which depend on the specific physical model being considered. Consequently, we can extend the preconditioning approach for an energy equation that can be expressed in the form of equation (17).

5 Preconditioning approach

From this point on, we are going to follow the derivation made in [1]. First, we can write the Navier-Stokes equations as:

$$\frac{\partial U}{\partial t} + \nabla \cdot \bar{F}^c (U) - \nabla \cdot \bar{F}^v (U, \nabla U) - Q = 0 \quad (19)$$

here, the conservative variables are $U = (\rho, \rho \vec{u}, \rho E)^T$. The source term, convective and viscous fluxes are given as follows:

$$\bar{F}^c (U) = \begin{pmatrix} \rho \vec{u} \\ \rho \vec{u} \otimes \vec{u} + \mathbf{I} p \\ \rho E \vec{u} \end{pmatrix}, \quad \bar{F}^v (U, \nabla U) = \begin{pmatrix} \cdot \\ \underline{\tau} \\ F_M(T, \nabla T) \end{pmatrix}, \quad Q = \begin{pmatrix} \cdot \\ \cdot \\ Q_M(U) \end{pmatrix} \quad (20)$$

we consider the set of primitive variables $V = (p, \vec{v}, q)^T$, where q is the working variable to be solved for, which can be T , h_s or h . Subsequently, the Navier-Stokes (N-S) equations (19) are transformed from conservative to primitive variables:

$$M \frac{\partial V}{\partial t} + \nabla \cdot \bar{F}^c (V) - \nabla \cdot \bar{F}^v (V, \nabla V) - Q = 0 \quad (21)$$

where:

$$M = \frac{\partial U}{\partial V} = \begin{pmatrix} \rho_p & 0 & 0 & 0 & \rho_q \\ \rho_p u & \rho & 0 & 0 & \rho_q u \\ \rho_p v & 0 & \rho & 0 & \rho_q v \\ \rho_p w & 0 & 0 & \rho & \rho_q w \\ \rho_p E & 0 & 0 & 0 & \rho_q E + \rho E_q \end{pmatrix} \quad (22)$$

with:

$$\rho_p = \frac{\partial \rho}{\partial p}, \quad \rho_q = \frac{\partial \rho}{\partial q}, \quad E_q = \frac{\partial E}{\partial q} \quad (23)$$

using the chain rule, we have:

$$\frac{\partial \rho}{\partial q} = \frac{\partial \rho}{\partial T} \cdot \frac{\partial T}{\partial q}, \quad \frac{\partial E}{\partial q} = \frac{\partial E}{\partial T} \cdot \frac{\partial T}{\partial q} \quad (24)$$

and:

$$q = \begin{cases} T \\ h_s \\ h \end{cases} \quad (25)$$

It must be noted that, for multicomponent and reacting flows, we have:

$$E = h, q = h \implies \rho_h = \frac{\partial \rho}{\partial h} = \frac{\partial \rho}{\partial T} \cdot \frac{\partial T}{\partial h} = -\frac{\rho}{C_p T}, E_q = 1 \quad (26)$$

Similarly, we have:

$$\frac{\partial V}{\partial U} = \frac{\partial T}{\partial h} = \frac{1}{C_p} \quad (27)$$

the above expressions remain valid if we change h by h_s .

If we multiply (21) by :

$$K = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ -u & 1 & 0 & 0 & 0 \\ -v & 0 & 1 & 0 & 0 \\ -w & 0 & 0 & 1 & 0 \\ -\textcolor{red}{E} & 0 & 0 & 0 & 1 \end{pmatrix} \quad (28)$$

Then, we have:

$$KM \frac{\partial V}{\partial t} + K \nabla \cdot \bar{F}^c(V) - K \nabla \cdot \bar{F}^v(V, K \nabla V) - KQ = 0 \quad (29)$$

where the product KM is giving by:

$$KM = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ -u & 1 & 0 & 0 & 0 \\ -v & 0 & 1 & 0 & 0 \\ -w & 0 & 0 & 1 & 0 \\ -\textcolor{red}{E} & 0 & 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} \rho_p & 0 & 0 & 0 & \rho_q \\ \rho_p u & \rho & 0 & 0 & \rho_q u \\ \rho_p v & 0 & \rho & 0 & \rho_q v \\ \rho_p w & 0 & 0 & \rho & \rho_q w \\ \rho_p \textcolor{red}{E} & 0 & 0 & 0 & \rho_q \textcolor{red}{E} + \rho E_q \end{pmatrix} = \begin{pmatrix} \rho_p & 0 & 0 & 0 & \rho_q \\ 0 & \rho & 0 & 0 & 0 \\ 0 & 0 & \rho & 0 & 0 \\ 0 & 0 & 0 & \rho & 0 \\ 0 & 0 & 0 & 0 & \rho \textcolor{red}{E}_q \end{pmatrix} \quad (30)$$

From here, we can define the preconditioning matrix as:

$$\Gamma_{nc} = \begin{pmatrix} \frac{1}{\beta^2} & 0 & 0 & 0 & \rho_q \\ 0 & \rho & 0 & 0 & 0 \\ 0 & 0 & \rho & 0 & 0 \\ 0 & 0 & 0 & \rho & 0 \\ 0 & 0 & 0 & 0 & \rho \textcolor{red}{E}_q \end{pmatrix} \quad (31)$$

where, following the choosing made by [1], we use:

$$\beta^2 = \epsilon (\vec{v} \cdot \vec{v})_{\max} \quad (32)$$

where ϵ is a user-provided value set as 2.0 by default, and $(\vec{v} \cdot \vec{v})_{\max}$ is the maximum squared velocity in the domain. Then, multiplying (29) by:

$$K^{-1} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ u & 1 & 0 & 0 & 0 \\ v & 0 & 1 & 0 & 0 \\ w & 0 & 0 & 1 & 0 \\ \textcolor{red}{E} & 0 & 0 & 0 & 1 \end{pmatrix} \quad (33)$$

we obtain:

$$\Gamma \frac{\partial V}{\partial t} + \nabla \cdot \bar{F}^c(V) - \nabla \cdot \bar{F}^v(V, \nabla V) - Q = 0 \quad (34)$$

where:

$$\Gamma = \begin{pmatrix} \frac{1}{\beta^2} & 0 & 0 & 0 & \rho_q \\ \frac{u}{\beta^2} & \rho & 0 & 0 & \rho_q u \\ \frac{v}{\beta^2} & 0 & \rho & 0 & \rho_q v \\ \frac{w}{\beta^2} & 0 & 0 & \rho & \rho_q w \\ \frac{E}{\beta^2} & 0 & 0 & 0 & \rho_q E + \rho E_q \end{pmatrix} \quad (35)$$

and:

$$\Gamma^{-1} = \begin{pmatrix} \frac{\beta^2 \rho_q E}{\rho E_q} + \beta^2 & 0 & 0 & 0 & -\frac{\beta^2 \rho_q}{\rho E_q} \\ -\frac{u}{\rho} & \frac{1}{\rho} & 0 & 0 & 0 \\ -\frac{v}{\rho} & 0 & \frac{1}{\rho} & 0 & 0 \\ -\frac{w}{\rho} & 0 & 0 & \frac{1}{\rho} & 0 \\ -\frac{E}{\rho E_q} & 0 & 0 & 0 & \frac{1}{\rho E_q} \end{pmatrix} \quad (36)$$

Finally, the projected convective flux is given by:

$$\overline{F}^c \cdot \hat{n} = \begin{pmatrix} \rho (\vec{u} \cdot \hat{n}) \\ \rho u (\vec{u} \cdot \hat{n}) + p n_x \\ \rho v (\vec{u} \cdot \hat{n}) + p n_y \\ \rho w (\vec{u} \cdot \hat{n}) + p n_z \\ \rho E (\vec{u} \cdot \hat{n}) \end{pmatrix} \quad (37)$$

where $\hat{n} = (n_x, n_y, n_z)$ is the local normal unit vector.

Then, computing the projected convective flux jacobian:

$$\frac{\partial (\overline{F}^c \cdot \hat{n})}{\partial V} = \begin{pmatrix} \rho_p (\vec{u} \cdot \hat{n}) & \rho n_x & \rho n_y & \rho n_z & \rho_q (\vec{u} \cdot \hat{n}) \\ \rho_p u (\vec{u} \cdot \hat{n}) + n_x & \rho (\vec{u} \cdot \hat{n}) + \rho u n_x & \rho u n_y & \rho u n_z & \rho_q u (\vec{u} \cdot \hat{n}) \\ \rho_p v (\vec{u} \cdot \hat{n}) + n_y & \rho v n_x & \rho (\vec{u} \cdot \hat{n}) + \rho v n_y & \rho v n_z & \rho_q v (\vec{u} \cdot \hat{n}) \\ \rho_p w (\vec{u} \cdot \hat{n}) + n_z & \rho w n_x & \rho w n_y & \rho (\vec{u} \cdot \hat{n}) + \rho w n_z & \rho_q w (\vec{u} \cdot \hat{n}) \\ \rho_p E (\vec{u} \cdot \hat{n}) & \rho E n_x & \rho E n_y & \rho E n_z & \rho_q E (\vec{u} \cdot \hat{n}) + \rho E_q (\vec{u} \cdot \hat{n}) \end{pmatrix} \quad (38)$$

Noting that $\rho_p = \frac{1}{\beta^2}$, we have:

$$\Gamma^{-1} \cdot \frac{\partial (\overline{F}^c \cdot \hat{n})}{\partial V} = \begin{pmatrix} \frac{(\vec{u} \cdot \hat{n})}{n_x} & \beta^2 \rho n_x & \beta^2 \rho n_y & \beta^2 \rho n_z & 0 \\ \frac{\rho}{n_y} & (\vec{u} \cdot \hat{n}) & 0 & 0 & 0 \\ \frac{\rho}{n_z} & 0 & (\vec{u} \cdot \hat{n}) & 0 & 0 \\ \frac{\rho}{0} & 0 & 0 & (\vec{u} \cdot \hat{n}) & 0 \\ 0 & 0 & 0 & 0 & (\vec{u} \cdot \hat{n}) \end{pmatrix} \quad (39)$$

Furthermore, computing the eigenvalues of the preconditioning system (for more details, see 14.2), we obtain:

$$\Lambda_\Gamma = \lambda (\overline{A}_\Gamma^c) = \lambda \left(\Gamma^{-1} \cdot \frac{\partial (\overline{F}^c \cdot \hat{n})}{\partial V} \right) = \begin{pmatrix} \vec{u} \cdot \hat{n} & 0 & 0 & 0 & 0 \\ 0 & \vec{u} \cdot \hat{n} & 0 & 0 & 0 \\ 0 & 0 & \vec{u} \cdot \hat{n} & 0 & 0 \\ 0 & 0 & 0 & (\vec{u} \cdot \hat{n}) - \beta & 0 \\ 0 & 0 & 0 & 0 & (\vec{u} \cdot \hat{n}) + \beta \end{pmatrix} \quad (40)$$

Hence, we obtain the same eigenvalues as in [1], and we can conclude that β acts as a pseudo-speed of sound as was concluded in [1].

6 Heat flux

Depending on the particular model, the heat flux takes the form given as follows:

$$F_M(T, \nabla T) = \begin{pmatrix} \kappa \nabla T & , & \text{perfect gas} \\ \kappa \nabla T + \rho \sum_{i=1}^N h_{s,i} D_i \nabla Y_i & , & \text{real gas sensible enthalpy} \\ \kappa \nabla T + \rho \sum_{i=1}^N h_i D_i \nabla Y_i & , & \text{real gas enthalpy} \end{pmatrix} \quad (41)$$

7 Source Term

Similarly, depending on the particular model, we have the following source term:

$$Q_M(U) = \begin{cases} \cdot & , & \text{perfect gas} \\ \dot{w}_T & , & \text{real gas sensible enthalpy} \\ \cdot & , & \text{real gas enthalpy} \end{cases} \quad (42)$$

8 Heat flux Jacobian

From [4], we have the computation of the heat flux jacobian given by:

$$\frac{\partial \bar{F}^v(U, \nabla U)}{\partial U} = \frac{\partial \bar{F}^v(V, \nabla V)}{\partial V} \cdot \frac{\partial V}{\partial U} \quad (43)$$

For the heat flux in the energy equation (8), we have $U = T, h_s$, or h , and $V = T$. Then, we must compute the following gradient:

$$\frac{\partial}{\partial T} \left(\kappa \nabla T + \rho \sum_{i=1}^N h_i D_i \nabla Y_i \right) = \frac{\partial}{\partial T} (\kappa \nabla T) + \frac{\partial}{\partial T} \left(\rho \sum_{i=1}^N h_i D_i \nabla Y_i \right) \quad (44)$$

For the second term in the right-hand side, we have:

$$\frac{\partial}{\partial T} \left(\rho \sum_{i=1}^N h_i D_i \nabla Y_i \right) = \rho \left(\sum_{i=1}^N \frac{\partial h_i}{\partial T} D_i \nabla Y_i \right) \quad (45)$$

where it is assumed that ρD_i remains constant, which is usually not true. Moreover, from (4), we obtain:

$$\frac{\partial h_i}{\partial T} = C_{p,i} \quad (46)$$

Finally, as the last species Y_N is not solved directly from the species transport equation but computed from mass conservation

$$Y_N = 1 - \sum_{i=1}^{N-1} Y_i \quad (47)$$

then, it follows that:

$$\nabla Y_N = - \sum_{i=1}^{N-1} \nabla Y_i \quad (48)$$

Hence, (45) can be expressed as:

$$\frac{\partial}{\partial T} \left(\rho \sum_{i=1}^N h_i D_i \nabla Y_i \right) = \rho \underbrace{\left(\sum_{i=1}^{N-1} (C_{p,i} D_i - C_{p,N} D_N) \nabla Y_i \right)}_{\vec{\psi}} \quad (49)$$

where $\vec{\psi}$ is a vector computed in each cell and its normal product with the normal vector to the cell face must be added to the solver.

9 Axisymmetric flows

For axisymmetric flows, it is possible to rewrite the equations in a cylindrical coordinate system such as we obtain a 2D formulation of the equations that can be solved numerically for 2D planar, and axisymmetric flows adding a contribution to the source term given by:

$$Q_{\text{add}} = \begin{pmatrix} -\rho \frac{v}{y} \\ -\rho u \frac{v}{y} + \frac{\mu}{y} \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) - \frac{2}{3} \frac{\partial}{\partial x} \left(\mu \frac{v}{y} \right) \\ -\rho v \frac{v}{y} + 2 \frac{\mu}{y} \left(\frac{\partial v}{\partial y} - \frac{v}{y} \right) - \frac{2}{3} \frac{\partial}{\partial y} \left(\mu \frac{v}{y} \right) \\ -\rho E \frac{v}{y} + \frac{\kappa}{y} \frac{\partial T}{\partial y} + \frac{\rho}{y} \sum_{i=1}^N h_i D_i \frac{\partial Y_i}{\partial y} \end{pmatrix} \quad (50)$$

hence, the total source term for axisymmetric flows becomes:

$$Q_T = Q + Q_{\text{add}} \quad (51)$$

Note that the term in blue in (50) must be added for multicomponent and reacting flows. It must also be noted from (41) that the blue term is also valid if we substitute h_i by $h_{s,i}$. Due to gradients of enthalpy, the following contribution to the jacobian is added in the energy equation:

$$\frac{\partial Q_E}{\partial h} = -\frac{\rho v}{y} \left(1 - \frac{h}{C_p T} \right) + \frac{\rho}{y C_p} \sum_{i=1}^N c_i D_i \frac{\partial Y_i}{\partial y} \quad (52)$$

the above expression remains valid when h_s is used as working variable.

10 Wall thermal boundary conditions

For isothermal walls, the heat flux at the fluid side is computed by:

$$q = k \frac{\partial T}{\partial n} \Big|_{\text{wall}} \quad (53)$$

where the gradient is computed as the difference between the interior node and the wall node

$$\frac{\partial T}{\partial n} \Big|_{\text{wall}} = \frac{T_i - T_w}{|\vec{x}_3 - \vec{x}_1|} \quad (54)$$

The heat flux jacobian at the wall is computed as:

$$\frac{\partial q}{\partial h} \Big|_{\text{wall}} = \frac{\partial q}{\partial T} \Big|_{\text{wall}} \cdot \frac{\partial T}{\partial h} = \frac{k}{|\vec{x}_3 - \vec{x}_1|} \cdot \frac{1}{C_p} \quad (55)$$

the same remains valid when h_s is considered instead of h .

For fluid-solid interaction, conjugate heat transfer is used to match the heat fluxes at each side of the interface. A Direct-temperature-Neumann boundary conditions is employed where we have:

$$T_{\text{wall}} = T_s \quad , \quad \text{fluid side} \quad (56)$$

$$k_s \nabla q = q_f \quad , \quad \text{solid side} \quad (57)$$

where the subscripts s and f refer to solid and fluid, respectively.

11 Cantera: Computation thermochemical properties

The specific heat capacity at constant pressure is computed by (3). However, in Cantera library, the specific heat capacities at constant pressure of species i , $C_{p,i}$, cannot be obtained directly. The scaled molar heat capacities at constant pressure of species i , $C_{p,i}^m/R_u$, are stored in Cantera as a function of T , where $R_u = 8.314[\text{J/mole K}]$ is the universal gas constant. Then, $C_{p,i}$ is computed using the following relation:

$$C_{p,i} = \frac{R_u}{W_i} \cdot \frac{C_{p,i}^m}{R_u}, \quad (58)$$

where W_i is the molecular weight of species i .

Similarly, the enthalpy of species i (4) is obtained as follows:

$$h_i = \frac{R_u T}{W_i} \cdot \frac{h_i^m}{R_u T} \quad (59)$$

where h_i^m is the molar enthalpy of species i . The highlighted term is stored as a function of T in Cantera.

12 Chemical Source term $\dot{\omega}_i$

The reaction rate $\dot{\omega}_i$ in (13) is computed from cantera as follows:

$$\dot{\omega}_i = W_i \mathcal{Q}_i \quad (60)$$

where $\mathcal{Q}_i$ is the net progress rate of species i defined as:

$$\mathcal{Q}_i = \sum_{j=1}^M \nu_{ij} \left(K_{fj} \prod_{j=1}^N [X_i]^{v'_{ij}} - K_{rj} \prod_{j=1}^N [X_i]^{v''_{ij}} \right) \quad (61)$$

where v'_{ij} , v''_{ij} the molar stoichiometric coefficients of species i in reaction j [3], with M chemical reactions. Furthermore, K_{fj} and K_{rj} are the forward and reverse rates of reaction j , respectively [3]. The forward and reverse rates of reaction j are modelled using the empirical Arrhenius law[3], and they are computed as follows:

$$K_{fj} = A_{fj} T^{\beta_j} \exp \left(-\frac{E_j}{RT} \right) \quad (62)$$

$$K_{rj} = \frac{K_{fj}}{\left(\frac{p_a}{RT} \right)^{\sum_{i=1}^N \nu_{ij}} \exp \left(\frac{\Delta S_0^j}{R} - \frac{\Delta H_0^j}{RT} \right)} \quad (63)$$

It must be noted that $\mathcal{Q}_i$ is written in terms of molar concentrations given by:

$$[X_i] = \frac{\rho Y_i}{W_i} \quad (64)$$

13 Mass flux correction continuity equation due to species diffusivity

As the sum of species mass fraction Y_i must be 1, then we have:

$$\rho = \sum_{i=1}^n \rho Y_i \quad (65)$$

hence, summing up the transport equations for species i (13), we have:

$$\sum_{i=1}^n \left[\frac{\partial(\rho Y_i)}{\partial t} + \nabla \cdot (\rho \vec{u} Y_i) \right] = \sum_{i=1}^n \nabla \cdot (\rho D_i \nabla Y_i) + \underbrace{\sum_{i=1}^n \dot{\omega}_i}_{=0} \quad (66)$$

$$\iff \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{u}) = \sum_{i=1}^{n-1} \nabla \cdot (\rho (D_i - D_N)) \nabla Y_i \quad (67)$$

which is the continuity equation equation that must be solved where diffusion effects are important (preferential diffusion for hydrogen simulations where the unity Lewis number assumption is not longer valid [2, 3]). It should be noted that the right-hand side of (67) becomes zero when the mass diffusivities are assumed to be equal for each species mass fraction Y_i .

14 Appendix

14.1 Eigenvalues Computation

We compute the characteristic polynomial of $\overline{A}_\Gamma^c$ as follows:

$$p_{\overline{A}_\Gamma^c}(\lambda) = \left| \overline{A}_\Gamma^c - \lambda \mathbf{I} \right| \quad (68)$$

$$= \begin{vmatrix} (\vec{u} \cdot \hat{n}) - \lambda & \beta^2 \rho n_x & \beta^2 \rho n_y & \beta^2 \rho n_z & 0 \\ \frac{n_x}{\rho} & (\vec{u} \cdot \hat{n}) - \lambda & 0 & 0 & 0 \\ \frac{n_y}{\rho} & 0 & (\vec{u} \cdot \hat{n}) - \lambda & 0 & 0 \\ \frac{n_z}{\rho} & 0 & 0 & (\vec{u} \cdot \hat{n}) - \lambda & 0 \\ 0 & 0 & 0 & 0 & (\vec{u} \cdot \hat{n}) - \lambda \end{vmatrix} \quad (69)$$

$$= ((\vec{u} \cdot \hat{n}) - \lambda) \begin{vmatrix} (\vec{u} \cdot \hat{n}) - \lambda & \beta^2 \rho n_x & \beta^2 \rho n_y & \beta^2 \rho n_z \\ \frac{n_x}{\rho} & (\vec{u} \cdot \hat{n}) - \lambda & 0 & 0 \\ \frac{n_y}{\rho} & 0 & (\vec{u} \cdot \hat{n}) - \lambda & 0 \\ \frac{n_z}{\rho} & 0 & 0 & (\vec{u} \cdot \hat{n}) - \lambda \end{vmatrix} \quad (70)$$

$$= ((\vec{u} \cdot \hat{n}) - \lambda) \begin{vmatrix} (\vec{u} \cdot \hat{n}) - \lambda - \frac{n_z^2 \beta^2}{(\vec{u} \cdot \hat{n}) - \lambda} & \beta^2 \rho n_x & \beta^2 \rho n_y & 0 \\ \frac{n_x}{\rho} & (\vec{u} \cdot \hat{n}) - \lambda & 0 & 0 \\ \frac{n_y}{\rho} & 0 & (\vec{u} \cdot \hat{n}) - \lambda & 0 \\ \frac{n_z}{\rho} & 0 & 0 & (\vec{u} \cdot \hat{n}) - \lambda \end{vmatrix} \quad (71)$$

$$= ((\vec{u} \cdot \hat{n}) - \lambda) ((\vec{u} \cdot \hat{n}) - \lambda) \begin{vmatrix} (\vec{u} \cdot \hat{n}) - \lambda - \frac{n_z^2 \beta^2}{(\vec{u} \cdot \hat{n}) - \lambda} & \beta^2 \rho n_x & \beta^2 \rho n_y \\ \frac{n_x}{\rho} & (\vec{u} \cdot \hat{n}) - \lambda & 0 \\ \frac{n_y}{\rho} & 0 & (\vec{u} \cdot \hat{n}) - \lambda \end{vmatrix} \quad (72)$$

$$= ((\vec{u} \cdot \hat{n}) - \lambda)^2 \begin{vmatrix} (\vec{u} \cdot \hat{n}) - \lambda - \frac{n_z^2 \beta^2}{(\vec{u} \cdot \hat{n}) - \lambda} - \frac{n_y^2 \beta^2}{(\vec{u} \cdot \hat{n}) - \lambda} & \beta^2 \rho n_x & 0 \\ \frac{n_x}{\rho} & (\vec{u} \cdot \hat{n}) - \lambda & 0 \\ \frac{n_y}{\rho} & 0 & (\vec{u} \cdot \hat{n}) - \lambda \end{vmatrix} \quad (73)$$

$$= ((\vec{u} \cdot \hat{n}) - \lambda)^3 \begin{vmatrix} (\vec{u} \cdot \hat{n}) - \lambda - \frac{n_z^2 \beta^2}{(\vec{u} \cdot \hat{n}) - \lambda} - \frac{n_y^2 \beta^2}{(\vec{u} \cdot \hat{n}) - \lambda} & \beta^2 \rho n_x \\ \frac{n_x}{\rho} & (\vec{u} \cdot \hat{n}) - \lambda \end{vmatrix} \quad (74)$$

$$= ((\vec{u} \cdot \hat{n}) - \lambda)^3 \cdot \left[\left((\vec{u} \cdot \hat{n}) - \lambda - \frac{n_z^2 \beta^2}{(\vec{u} \cdot \hat{n}) - \lambda} - \frac{n_y^2 \beta^2}{(\vec{u} \cdot \hat{n}) - \lambda} \right) \cdot ((\vec{u} \cdot \hat{n}) - \lambda) - \frac{n_x}{\rho} \cdot \beta^2 \rho n_x \right] \quad (75)$$

$$= ((\vec{u} \cdot \hat{n}) - \lambda)^3 \cdot \left[((\vec{u} \cdot \hat{n}) - \lambda)^2 - \underbrace{(n_x^2 + n_y^2 + n_z^2)}_{=1} \beta^2 \right] \quad (76)$$

$$= ((\vec{u} \cdot \hat{n}) - \lambda)^3 \cdot \left[((\vec{u} \cdot \hat{n}) - \lambda)^2 - \beta^2 \right] \quad (77)$$

$$= ((\vec{u} \cdot \hat{n}) - \lambda)^3 ((\vec{u} \cdot \hat{n}) - \lambda - \beta) ((\vec{u} \cdot \hat{n}) - \lambda + \beta) \quad (78)$$

Then, the eigenvalues are :

$$\lambda_1 = \lambda_2 = \lambda_3 = (\vec{u} \cdot \hat{n}) , \lambda_4 = (\vec{u} \cdot \hat{n}) - \beta , ; \lambda_5 = (\vec{u} \cdot \hat{n}) + \beta \quad (79)$$

14.2 Eigenvectors Computation

For each eigenvalues, we compute the eigenvector associated to them. First, we compute the characteristic space E_{λ_i} associated with each eigenvalue λ_i :

- $\lambda_1 = \vec{u} \cdot \hat{n}$, We have:

$$E_{\lambda_1} = \left\{ \vec{x} : \left(\bar{A}_\Gamma^c - \lambda_1 \mathbf{I} \right) \vec{X} = \vec{0} \right\} \quad (80)$$

$$= \left\{ \vec{x} : \begin{pmatrix} 0 & \beta^2 \rho n_x & \beta^2 \rho n_y & \beta^2 \rho n_z & 0 \\ \frac{n_x}{n_y} & 0 & 0 & 0 & 0 \\ \frac{\rho}{n_z} & 0 & 0 & 0 & 0 \\ \frac{\rho}{0} & 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \right\} \quad (81)$$

$$= \left\{ \vec{x} : \begin{pmatrix} \beta^2 \rho n_x x_2 + \beta^2 \rho n_y x_3 + \beta^2 \rho n_z x_4 = 0 \\ \frac{n_x}{n_y} x_1 = 0 \\ \frac{\rho}{n_z} x_1 = 0 \\ \frac{\rho}{0} x_1 = 0 \\ 0 \cdot x_5 = 0 \end{pmatrix} \right\} \quad (82)$$

$$= \left\{ \vec{x} : \begin{pmatrix} x_2 = -\frac{n_y}{n_x} x_3 - \frac{n_z}{n_x} x_4 \\ x_1 = 0 \\ x_5 \in \mathbb{R} \end{pmatrix} \right\} \quad (83)$$

$$= \left\{ \vec{x} : \vec{x} = \begin{pmatrix} 0 \\ -\frac{n_y}{n_x} x_3 - \frac{n_z}{n_x} x_4 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix}, x_3, x_4, x_5 \in \mathbb{R} \right\} \quad (84)$$

$$= \left\{ \vec{x} : \vec{x} = \begin{pmatrix} 0 \\ -\frac{n_y}{n_x} \\ 1 \\ 0 \\ 0 \end{pmatrix} x_3 + \begin{pmatrix} 0 \\ -\frac{n_z}{n_x} \\ 0 \\ 1 \\ 0 \end{pmatrix} x_4 + \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} x_5, x_3, x_4, x_5 \in \mathbb{R} \right\} \quad (85)$$

$$= \left\langle \left\{ \begin{pmatrix} 0 \\ -\frac{n_y}{n_x} \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ -\frac{n_z}{n_x} \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right\} \right\rangle \quad (86)$$

Hence, the eigenvectors associated to λ_1 are:

$$\vec{v}_1 = \begin{pmatrix} 0 \\ -\frac{n_y}{n_x} \\ 1 \\ 0 \\ 0 \end{pmatrix}, \vec{v}_2 = \begin{pmatrix} 0 \\ -\frac{n_z}{n_x} \\ 0 \\ 1 \\ 0 \end{pmatrix}, \vec{v}_3 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \quad (87)$$

Following an analogous procedure, we obtain the eigenvectors associated to the remaining eigenvalues.

- $\lambda_2 = \vec{u} \cdot \hat{n} - \beta$, We have:

$$\vec{v}_4 = \begin{pmatrix} -\frac{\beta\rho}{n_x} \\ \frac{n_z}{n_x} \\ \frac{n_z}{n_y} \\ n_z \\ 1 \\ 0 \end{pmatrix} \quad (88)$$

- $\lambda_3 = \vec{u} \cdot \hat{n} + \beta$, We have:

$$\vec{v}_5 = \begin{pmatrix} \frac{\beta\rho}{n_x} \\ \frac{n_z}{n_x} \\ \frac{n_z}{n_y} \\ n_z \\ 1 \\ 0 \end{pmatrix} \quad (89)$$

Hence, we construct the matrix P using the eigenvector as follows:

$$P = \begin{pmatrix} 0 & 0 & 0 & -\frac{\beta\rho}{n_x} & \frac{\beta\rho}{n_x} \\ -\frac{n_y}{n_x} & -\frac{n_z}{n_x} & 0 & \frac{n_z}{n_x} & \frac{n_z}{n_x} \\ \frac{n_y}{n_x} & \frac{n_z}{n_x} & 0 & \frac{n_z}{n_y} & \frac{n_z}{n_y} \\ 1 & 0 & 0 & n_z & n_z \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 \end{pmatrix} \quad (90)$$

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